

Paper II Audit -- P2: Vertex-copy inequality (Lemma 3.1)

Test	P2 -- Vertex-copy inequality (Lemma 3.1)
Gate	B
Evidence	EXHAUSTIVE (n<=5 all graphs) + random general/chordal (n=6..9)
Phi_tau method	auditor's own cover-LP (scipy HiGHS)
Master seed	20260710
Environment	Python 3.14.4 / networkx 3.6.1

1. Claim under test (Lemma 3.1)

For nonadjacent u, v : $\Phi_\tau(G_{\{v \rightarrow u\}}) + \Phi_\tau(G_{\{u \rightarrow v\}}) \geq 2\Phi_\tau(G)$, where the copy deletes all edges at v and re-joins v to $N(u)$. Corollary: at least one copy has $\Phi_\tau \geq \Phi_\tau(G)$. The lemma is general (no chordality, no $z_e \leq 1$).

2. Kill-switch (declared before running)

H2a the inequality; **H2b** the corollary; plus the exact edge-count identity $e(G_{\{v \rightarrow u\}}) + e(G_{\{u \rightarrow v\}}) = 2e(G)$. Any violation refutes the lemma.

3. Result

Graphs tested: **1576**; nonadjacent pairs: **11032**. Inequality violations: **0**. Corollary violations: **0**. Edge-identity failures: **0**. Tightest slack observed: **-0.000** (equality is achievable, slack 0).

Verdict: VERIFIED (in range): the vertex-copy inequality and its corollary hold on every tested (G, u, v) ; the edge-count identity holds exactly.

4. Scope

$n \leq 5$ is exhaustive over all labelled graphs (all nonadjacent pairs); $n=6..9$ mixes random general and random chordal graphs. The lemma is universal, so this is existential evidence -- but any single violation would have refuted it, and none was found. The exact edge-count identity (an algebraic step of the proof) holds in every case.